Exhaustive search and branch-and-bound

Week 5

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Recap

Weeks 3–4: overlapping subproblems and a local recurrence. A table then beats enumeration.

Some problems have no such overlap. The honest first method is to **try all candidates**, organised as a **tree**.

Restriction maps

Cut DNA at a known word. The list of those cuts was the first picture of a genome.

The first pictures of DNA were cut sites

Phage λ is 48.5 kb. In 1975 that was too long to sequence. EcoRI cut it at five sites. The **restriction map** — where those cuts sit — told you which fragment held a gene.



EcoRI cuts a six-letter word

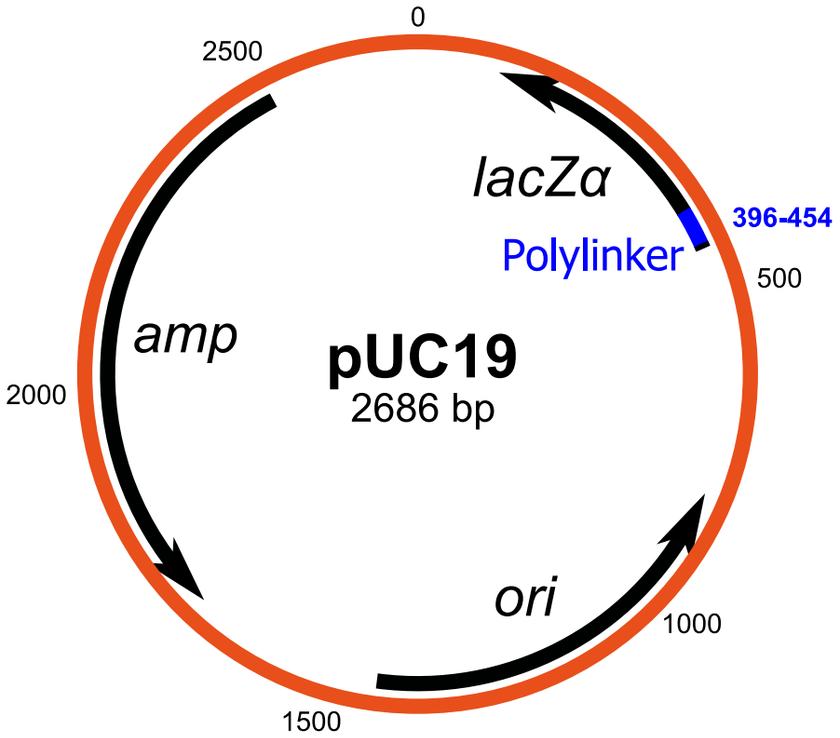
A restriction enzyme recognises a short DNA word and **cuts** there. EcoRI cuts `GAATTC`. Sticky ends let two molecules cut with the same enzyme join.



EcoRI recognition site and cut. Bryan Derksen, Wikimedia Commons, public domain. File: EcoRI restriction enzyme recognition site.svg.

A plasmid is still a restriction map

pUC19 is 2686 bp. Ampicillin (`amp`) and the origin (`ori`) must stay intact. EcoRI sits in the **polylinker**. Every cloning vector on Addgene is still drawn this way.



pUC19. Yikrazuul, Wikimedia Commons. Public domain (simple structural formula). File: PUC19.svg.

We sequence genomes. We still cut plasmids.

λ is a sequence now. You still open a plasmid at EcoRI. Restriction maps of *genomes* are history. Restriction *enzymes* are bench tools.

genomes, then



phage λ from gel bands

today we sequence it

plasmids, still

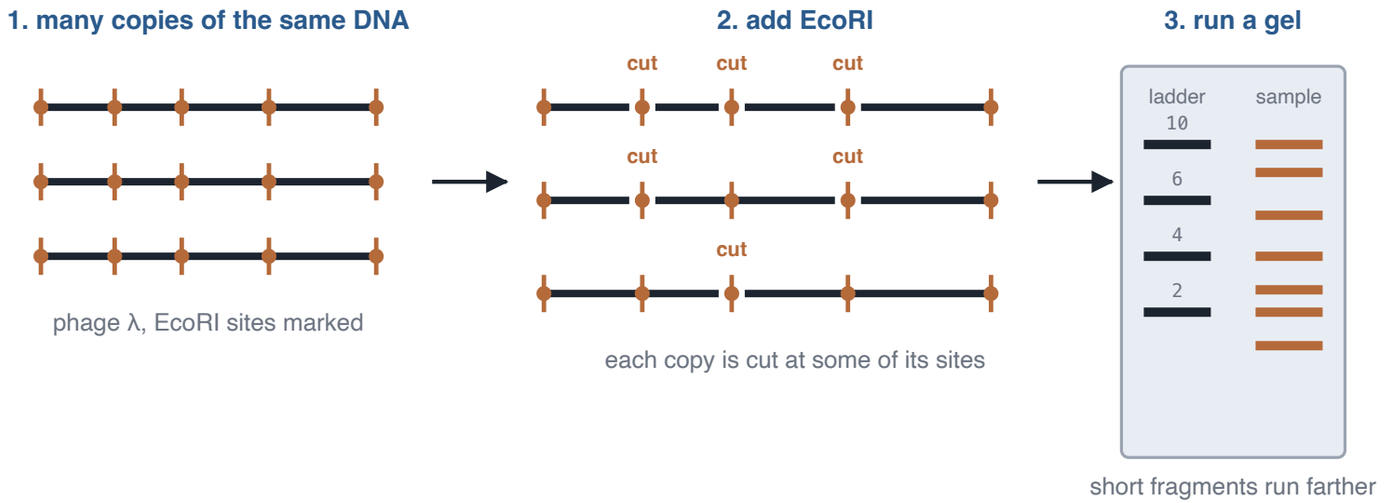


every cloning vector

Addgene, SnapGene, NEB

How they measured the map

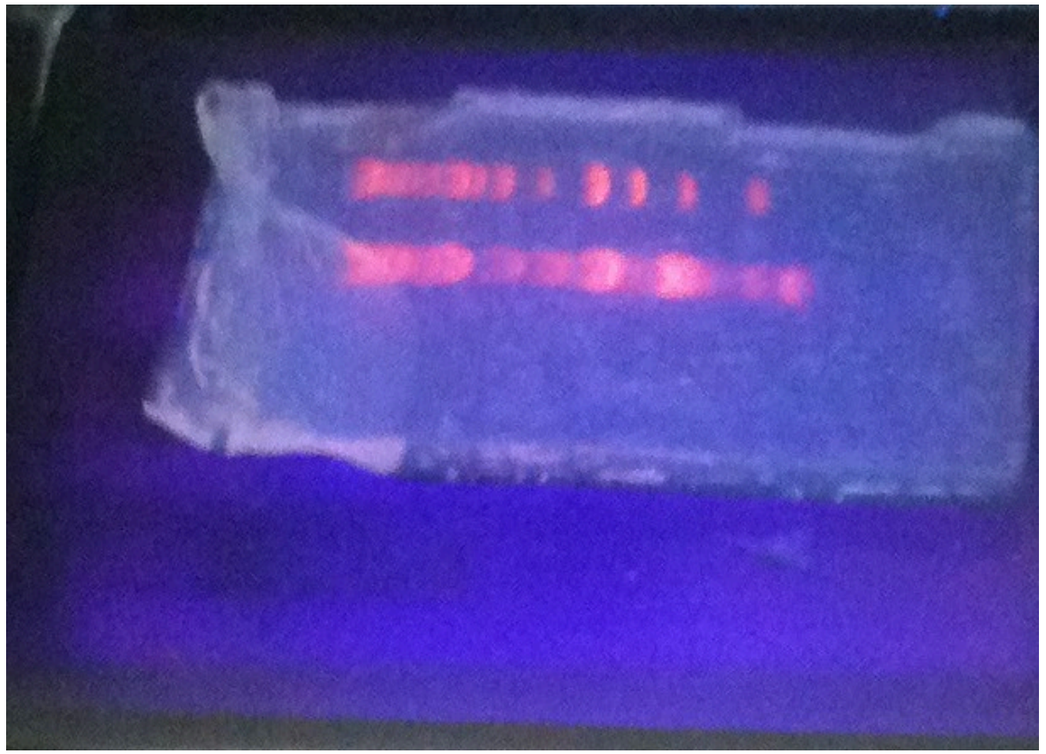
Many identical copies of the phage DNA. Add EcoRI. Each copy is cut at *some* of its `GAATTC` sites — you do not choose which. Load the mixture on a gel: short fragments run farther.



A ladder of known sizes turns each band into a length. Positions are still unknown.

The gel gives lengths

Each band is a fragment length. The cut positions are the unknown.



Agarose gel with DNA ladders (UV). Joseph Elsbernd, Wikimedia Commons, CC BY-SA 2.0. File: Agarose gel with DNA ladders on a UV transilluminator.jpg.

A complete digest loses the order

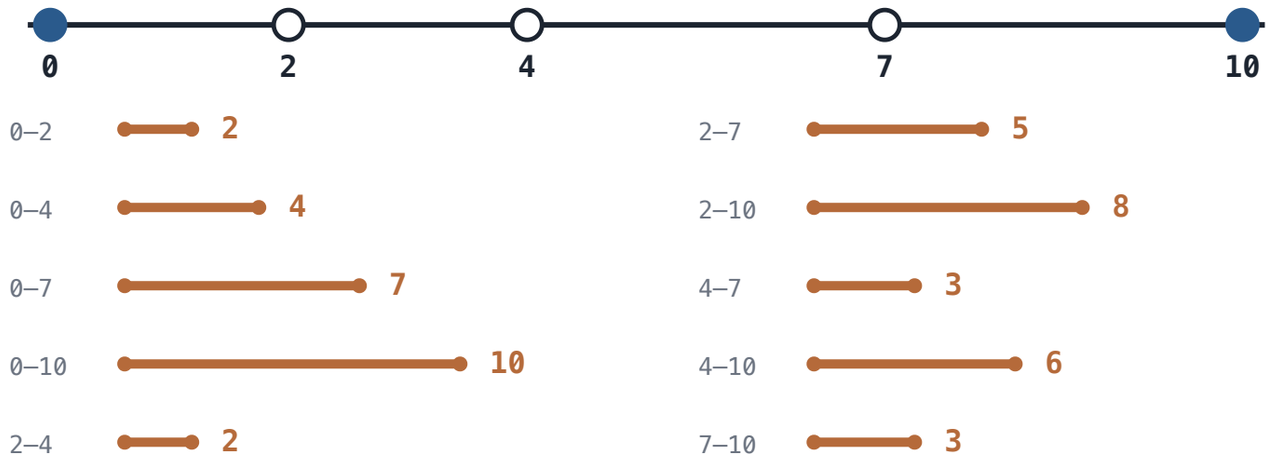
Wait long enough, or use enough enzyme, and almost every copy is cut at **every** site. The gel then shows only neighbouring pieces. Two fragments of length 2 and two of length 3 do not say which 2 sits next to which 3.



the gel shows 2, 2, 3, 3 — which 2 sits next to which 3 is lost

A partial digest lists every pair

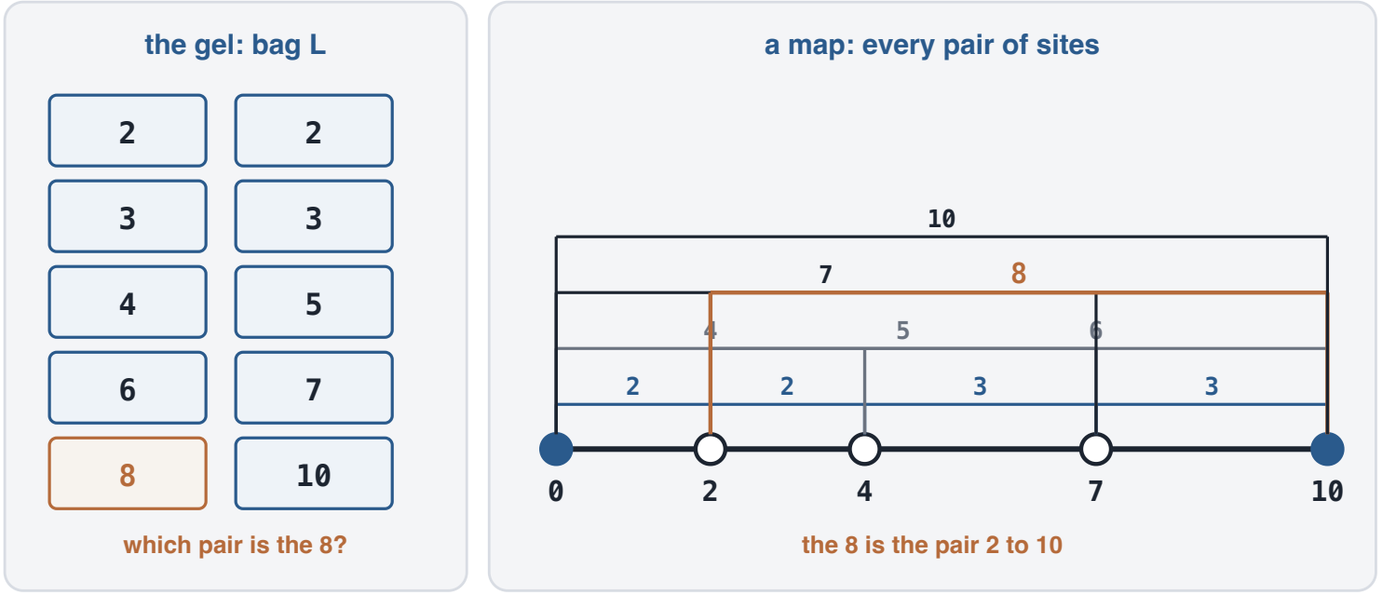
Stop early, or use less enzyme. Some copies are cut at two sites i and j with nothing cut in between, so that molecule is one fragment of length $\left|x_j - x_i\right|$. Across many copies, **every pair** of sites appears as a band.



Ten pairs, ten lengths. That bag is L.

The formulated problem

The gel gave an **unordered bag** L. Each length is $|x_i - x_j|$ for some pair of sites — neighbours, and every longer span. Lining the numbers up does not place the cuts: 8 might sit 8 from the left end, or 8 from the right. A map X is a solution only if **every** pairwise distance of X equals L.



Input / output

INPUT

A multiset L of pairwise distances among unknown points on a line.

→

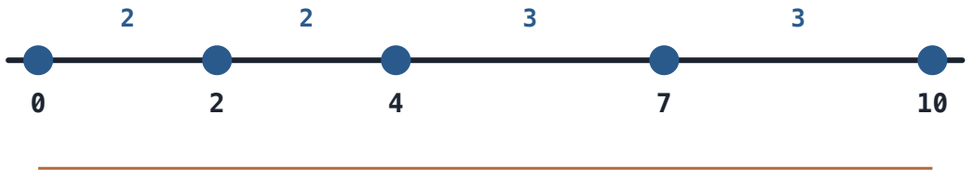
OUTPUT

Positions X (including the two ends) whose pairwise distances are exactly L.

Today's toy

Toy. Distances $L = \{2, 2, 3, 3, 4, 5, 6, 7, 8, 10\}$. Ends are 0 and 10. Recover $X = \{0, 2, 4, 7, 10\}$. Lab: `labs/05-search.ipynb`.

Points on a line. Input: all pairwise distances. Output: the positions.



largest distance = 10 => the ends are 0 and 10

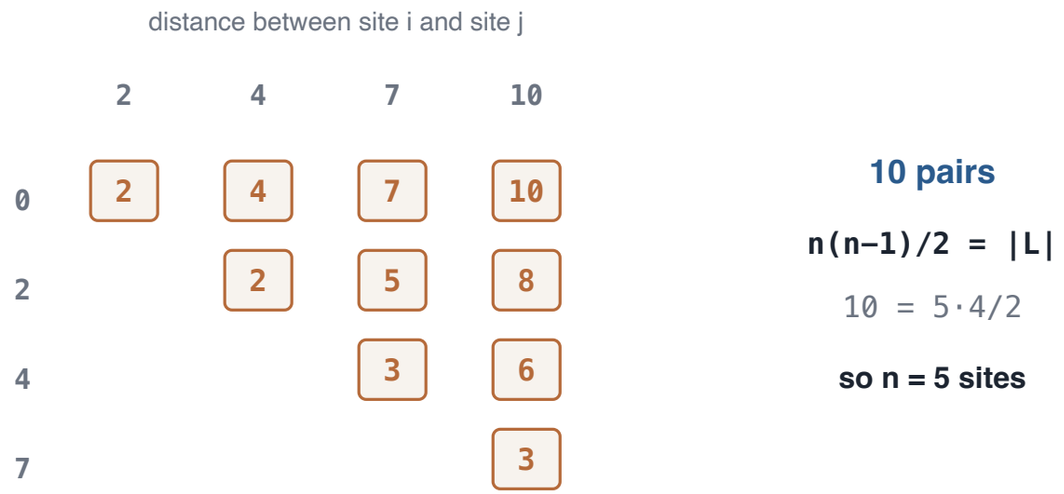
L = {2, 2, 3, 3, 4, 5, 6, 7, 8, 10} X = {0, 2, 4, 7, 10}

How many sites are on the map?

n sites produce one length for every pair of sites:

$$\binom{n}{2} = \frac{n(n-1)}{2} = |L|.$$

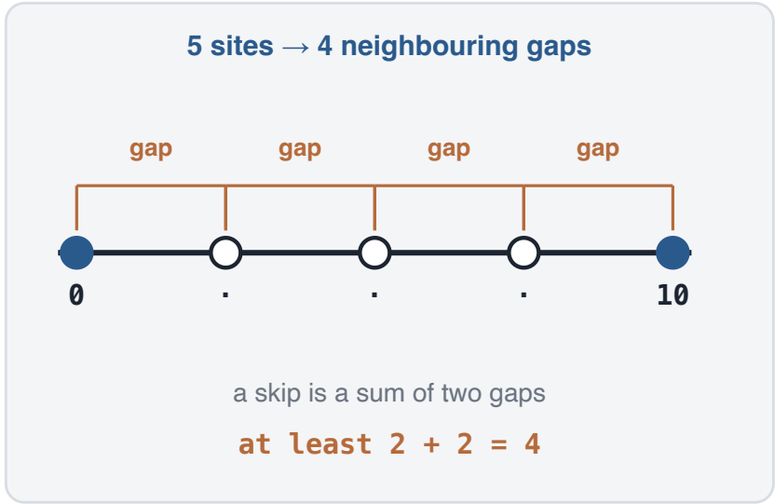
Here $|L| = 10$, so $n = 5$ (including the two ends).



Four smallest lengths, four gaps

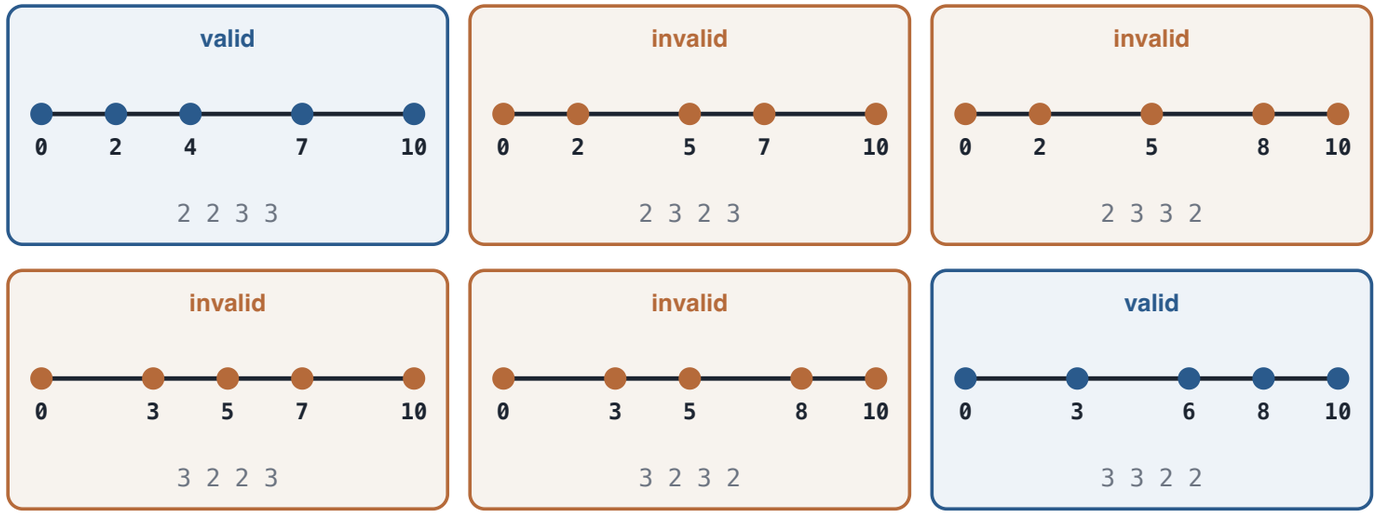
$n = 5$ sites sit on a line, so there are $n - 1 = 4$ neighbouring gaps. A longer pair is a sum of two or more gaps, hence at least $2 + 2 = 4$. The four smallest numbers in L — $2, 2, 3, 3$ — are those neighbouring distances. Their order is still unknown.





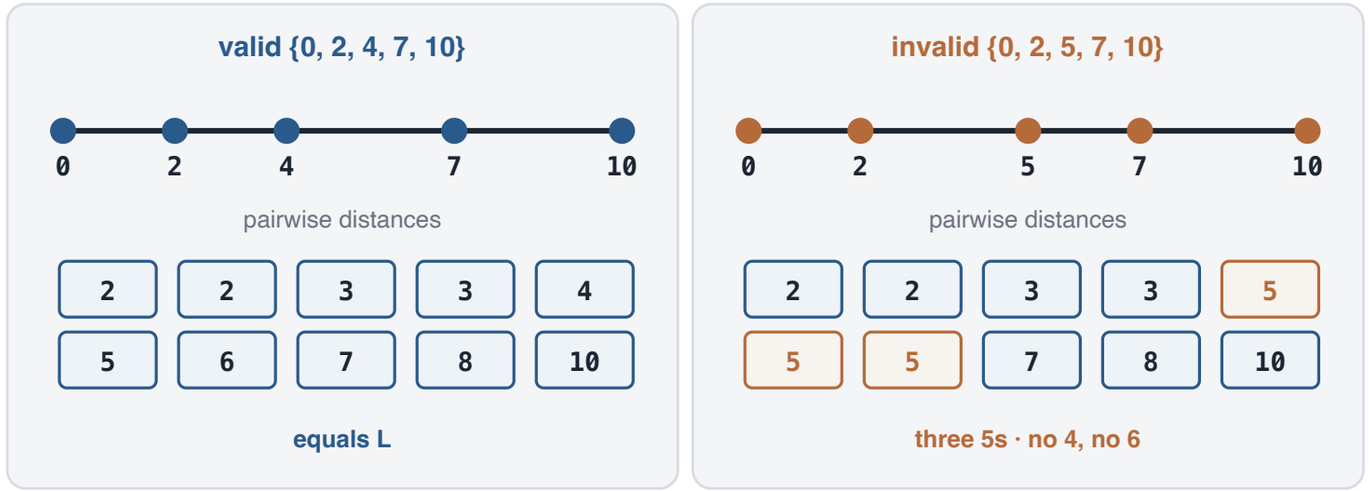
Six ways to line up $2, 2, 3, 3$

Line those four gaps up. Distinct orders: $4!/(2!\,2!) = 6$. Check each against the full bag L. Two maps match (a placement and its reverse). Four fail.



One valid, one invalid

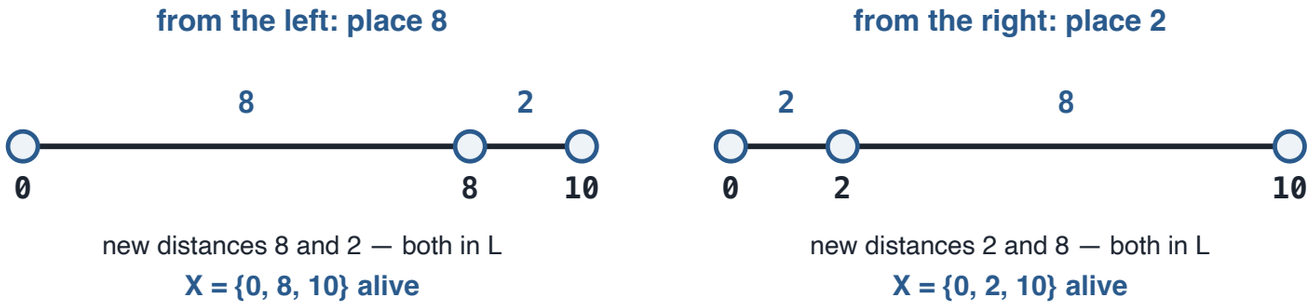
$X = \{0, 2, 4, 7, 10\}$ reproduces L. $X = \{0, 2, 5, 7, 10\}$ produces three 5s and no 4. Four of the six orders fail.



List the maps as a tree

The largest leftover length δ must sit δ from one of the two ends. Two children: place it from the left, or from the right. A node is a partial set X; an edge adds one site.

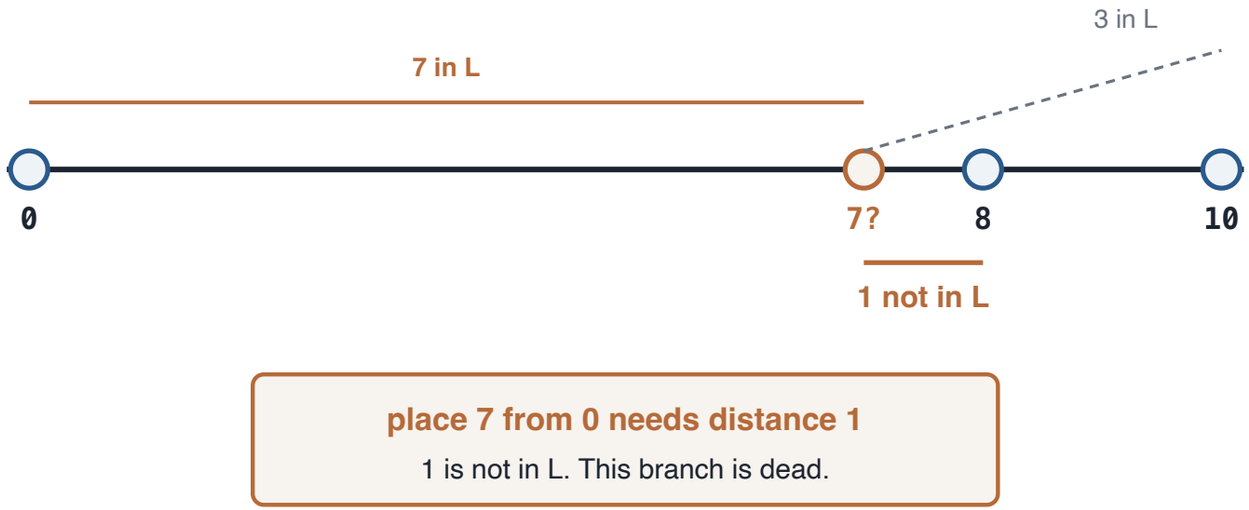
Start from $X = \{0, 10\}$. Largest leftover is 8. Both children live.



Largest leftover is 8. Both children live.

Some branches are illegal

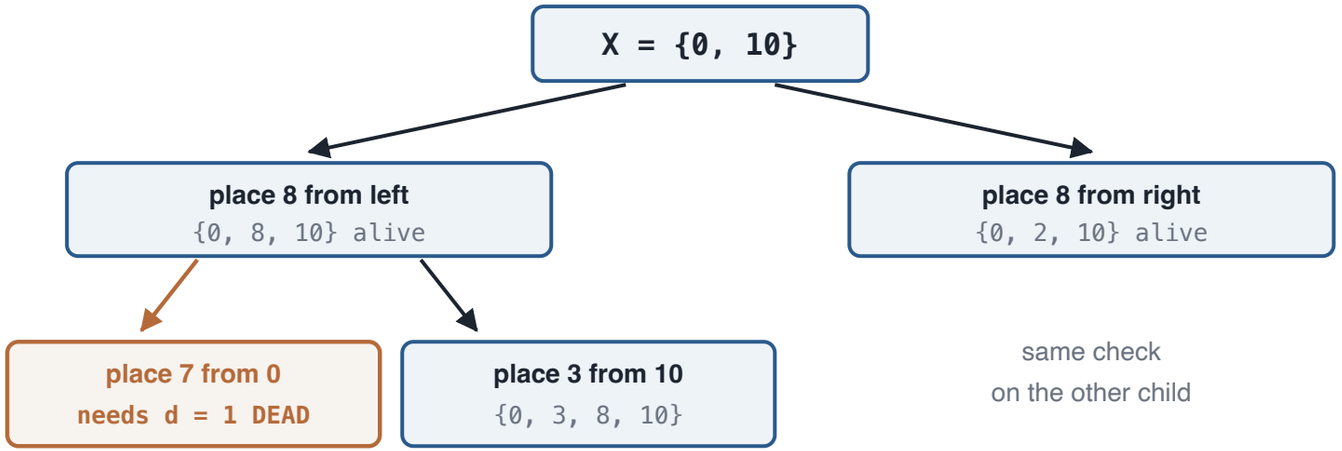
From $\{0, 8, 10\}$, leftover max is 7. Place 7 from the left: new distances 7, $|7 - 8| = 1$, and $|7 - 10| = 3$. The length 1 is not in the experimental bag L.



That map cannot be completed to match the gel. Stop the branch. The other child — place 3 from 10 — has $3, 5, 7$, all in L. Continue.

Skip that subtree

No completion of “place 7 from 0” can match L. One living branch later reaches $X = \{0, 2, 4, 7, 10\}$.



A missing length is a proof: no completion of this node can match L.

Skip the whole subtree.

The notebook checks that prune

`legal(X, x, L)` asks: do the new distances from x still sit in leftover L? If a length is missing, it returns that length.

```
1 legal([0, 8, 10], 7, L)   # (False, 1)  — illegal
2 legal([0, 8, 10], 3, L)   # (True, None)
```

In the lab: Run All, then uncomment `legal([0, 2, 10], 7, L)` — does that child live?

The algorithm, in steps

1. Start from $X = \{0, W\}$, where W is the longest length in L (the two ends of the molecule).
2. While leftover lengths remain, let δ be the largest unused length.
3. Two children: place a site δ from the left end, or δ from the right end.
4. A child lives only if every new pairwise distance still sits in leftover L. If a length is missing, skip that whole subtree.
5. Repeat. Each surviving leaf is a map whose pairwise distances are exactly L.

The algorithm is exact

A branch dies only when it already needs a length the gel did not produce. Every map whose pairwise distances *are* L is still a leaf of the remaining tree.

We skip work. We do not skip answers. The output is the same as walking every legal placement to the end.

That method is branch-and-bound

Walk a tree of partial solutions. At a node, a **bound** proves that no completion can succeed (here: a required length is missing). Skip the whole subtree.

> The skip is a proof about every descendant, so the search stays exact.

90 seconds: guess or proof?

Pair: is that bound a **guess**, or a **proof**?

If even the dream future of this node cannot match L, the real future cannot. Skip.

Cost of walking the whole tree

If every node has branching factor b and the depth is d, the tree has $\Theta\left(b^d\right)$ nodes. Illegal children cut that down.

> The search is a **binary tree of placements**. A missing length kills a branch. That skip is branch-and-bound.

Motifs

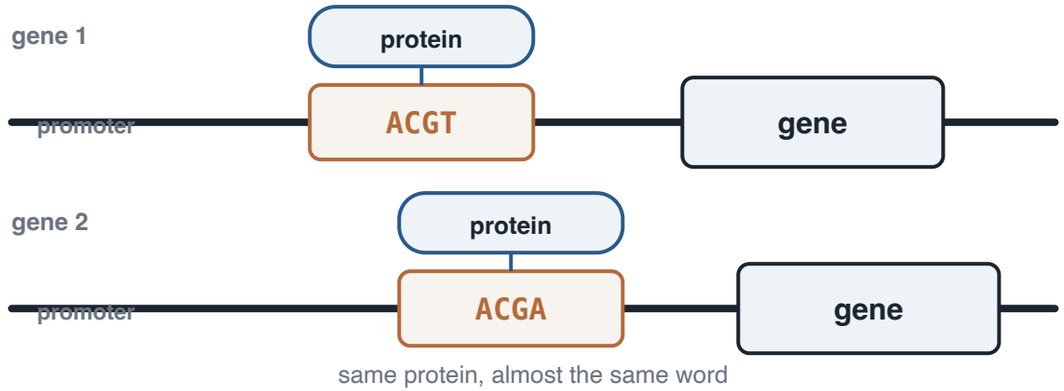
A short noisy word that turns genes on. Same kind of search tree. The bound will be a number.



TATA-box sequence logo. Ericbf10, Wikimedia Commons, CC BY-SA 4.0. File: Caja TATA.png.

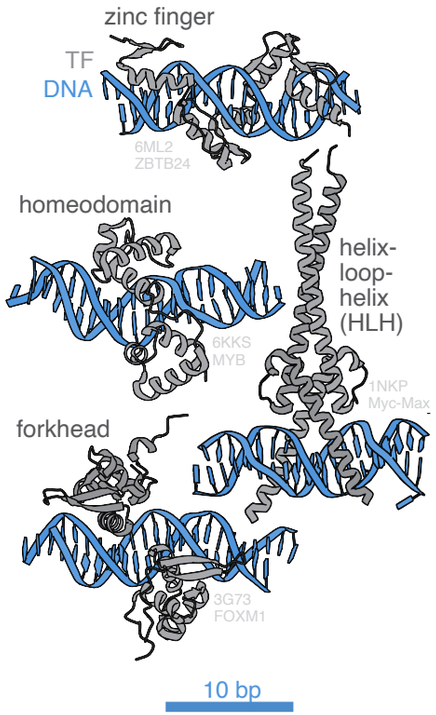
Why we care about motifs

A transcription factor binds a short word in the promoter and turns the gene on. Genes that fire **together** often share that word.



The protein contacts a handful of bases

The DNA-binding domain sits on a few base pairs — a zinc finger, a homeodomain, a helix–loop–helix. That is why the motif is short, and why a mismatch can still bind.

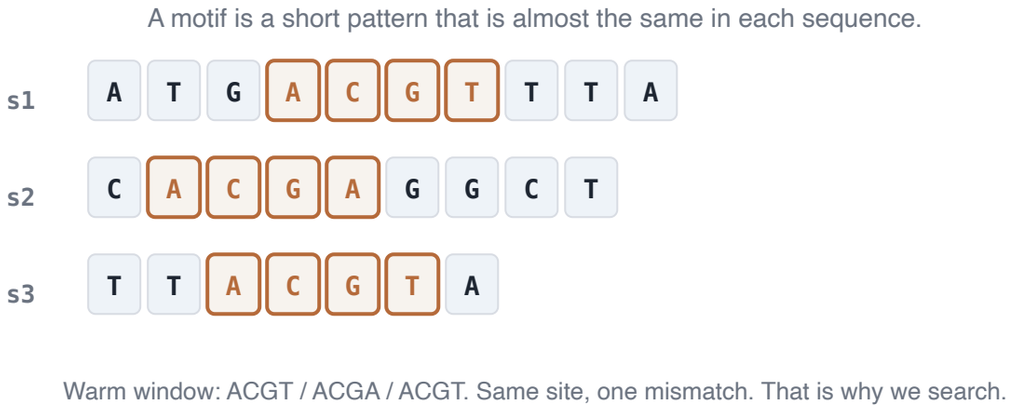


DNA-binding domains of transcription factors (zinc finger, homeodomain, helix–loop–helix, forkhead). J.J. Froehlich, Wikimedia Commons, CC BY-SA 4.0. File: Transcription factors DNA binding sites.svg.

The copies are not identical

The protein tolerates mismatches. The site in sequence 2 may be `ACGA` while the others are `ACGT`. One substitution, same biology.

If you take the promoters of co-expressed genes, the shared motif is a **hypothesis about who binds**.



The formulated problem

Find a short word that sits, with a few mismatches, in every sequence.

INPUT

t DNA strings (promoters) and a length k
.

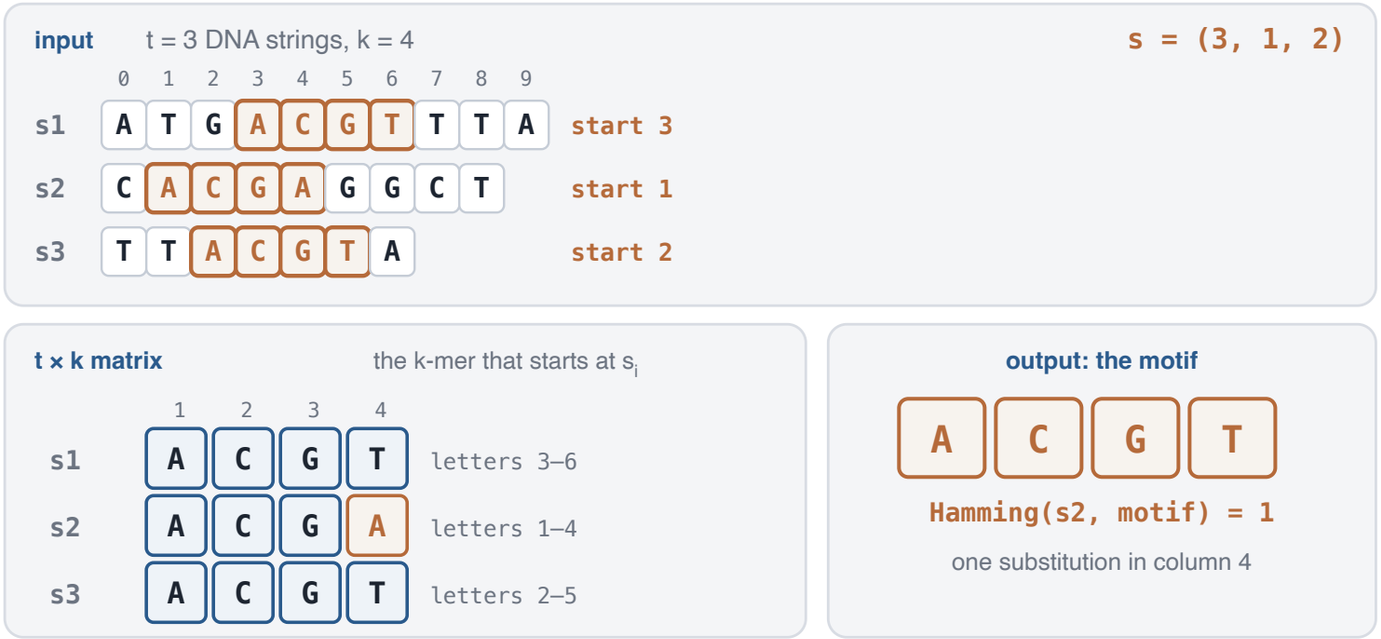
→

OUTPUT

Starting positions $s = (s_1, \ldots, s_t)$
and a motif of length k.

Input / output

Starting positions $s = (3, 1, 2)$: s_i is the start in string i. The k letters from each start form a $t \times k$ matrix. The motif is `ACGT`. **Hamming distance** against row s_2 is 1.



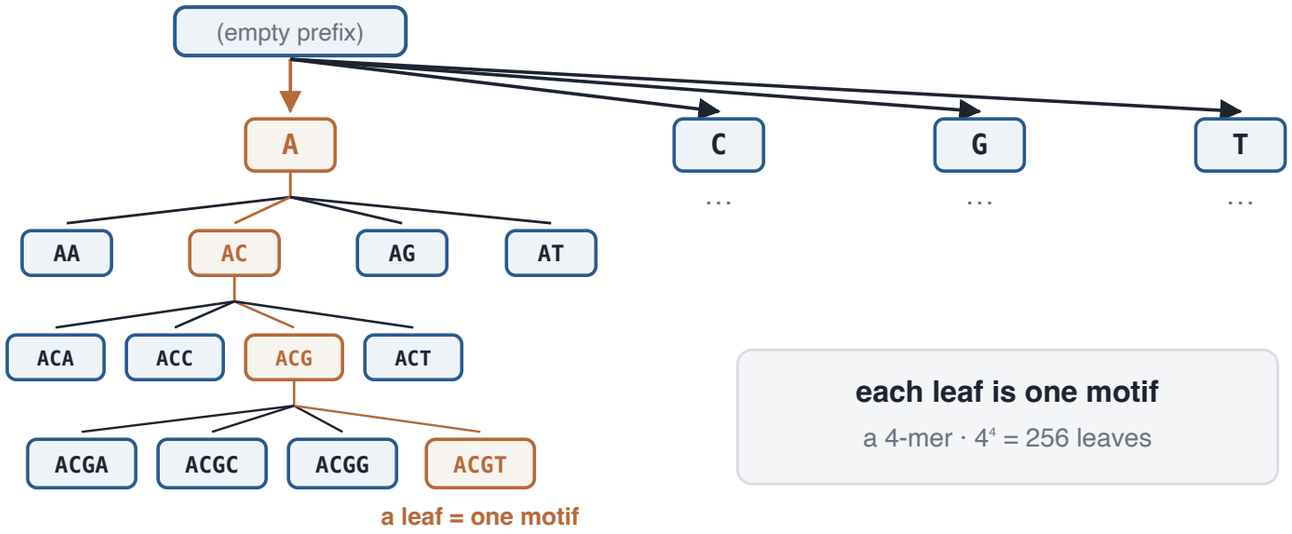
Hamming distance

Hamming distance counts substitutions between two words of the same length. Alignment in weeks 3–4 used **edit distance**: substitutions and gaps. On `GCT` versus `GCAT` that is one insertion.



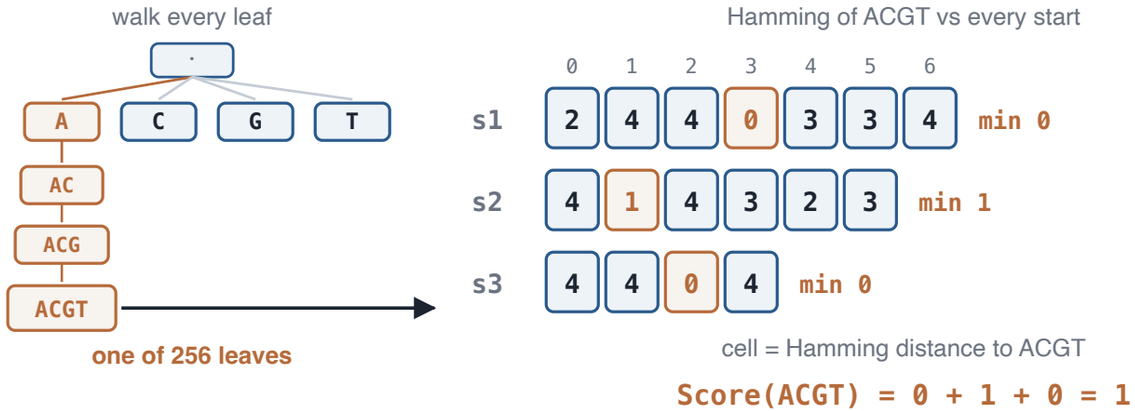
A tree whose leaves are motifs

Each edge adds one letter. Depth k. A leaf is one k-mer — one candidate motif. Branching factor 4. For $k = 4$ there are $4^4 = 256$ leaves.



Brute force: every motif, every start

Walk every leaf. At a leaf, Hamming that k-mer against **every** start; keep the closest window in each string and sum.

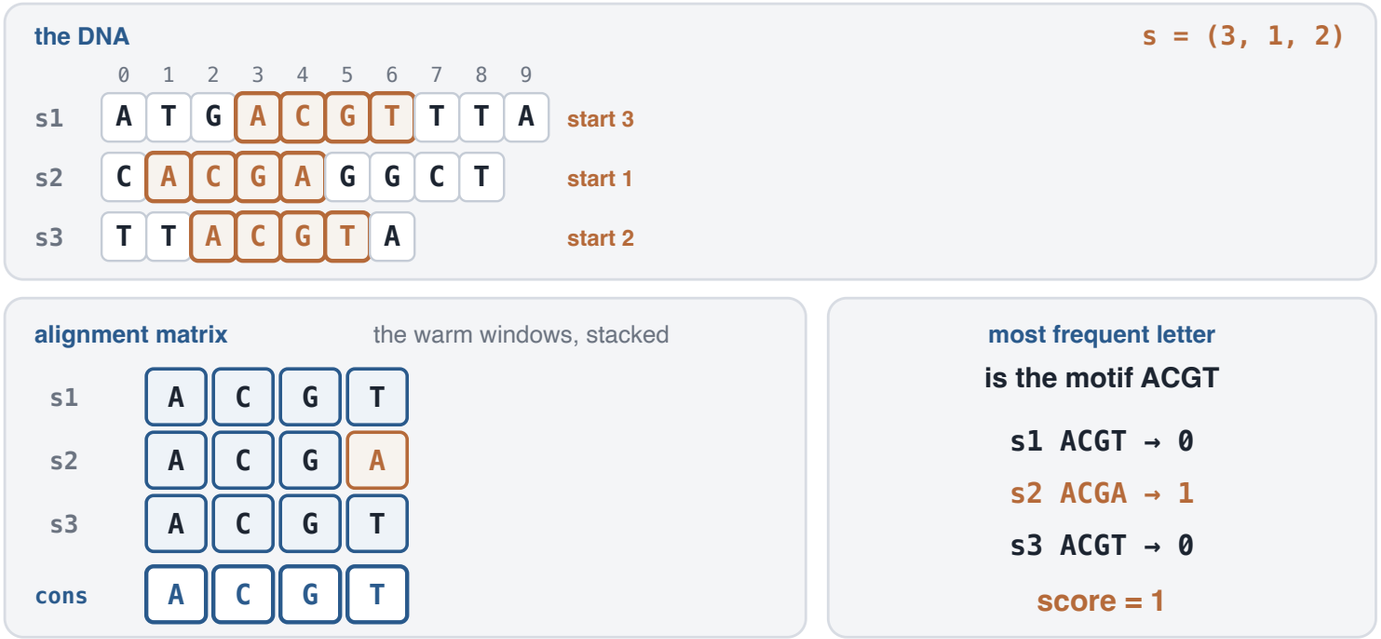


do this at every leaf: 256 motifs × 17 windows = 4352 tests

s1 has 10–4+1 = 7 starts. In all: 7 + 6 + 4 = 17.

Fix the starts instead

Choose one start in each sequence. The warm windows in the DNA **are** the alignment. Most frequent letter per column is the motif. Smaller score is better.

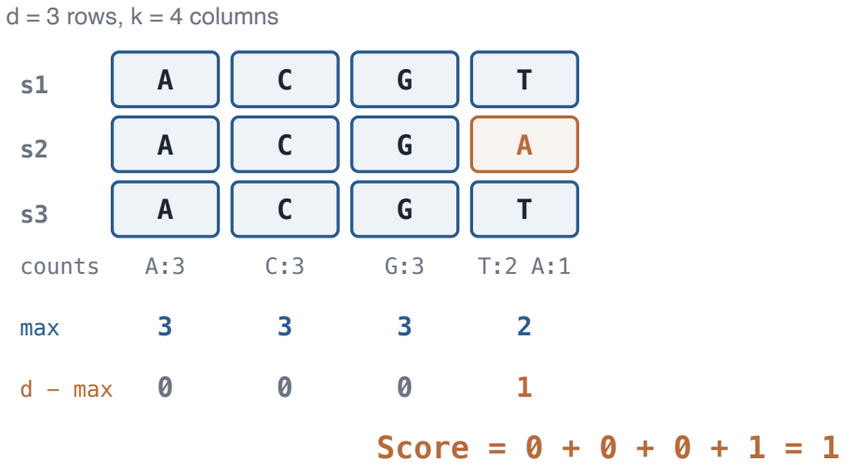


How the score is computed

$n_c(a)$ = how many of the d rows have letter a in column c. That column costs every letter that is not the majority.

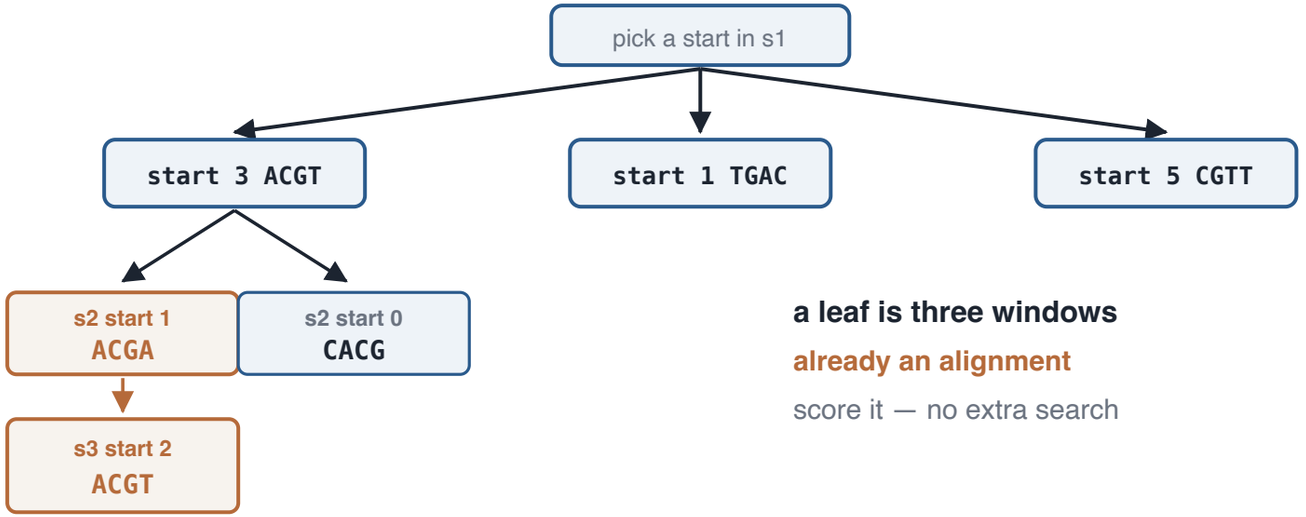
$$\text{Score} = \sum_{c=1}^{k} \left(d - \max_{a} n_c(a) \right)$$

On a complete alignment $d = t$. Smaller is better.



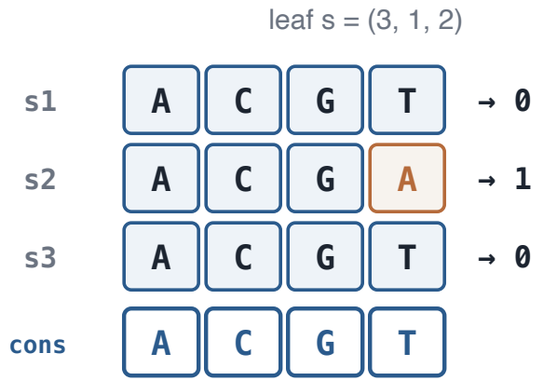
So the tree is over starting positions

Each edge picks a start in the next sequence. A leaf is already an alignment, so the consensus motif and its score come in one $O(tk)$ pass.



A complete alignment we will keep

The leaf $s = (3, 1, 2)$ is already an alignment. Consensus `ACGT`, one mismatch in s_2. Score $= 1$. Call that best-so-far $B = 1$.



consensus score B = 1

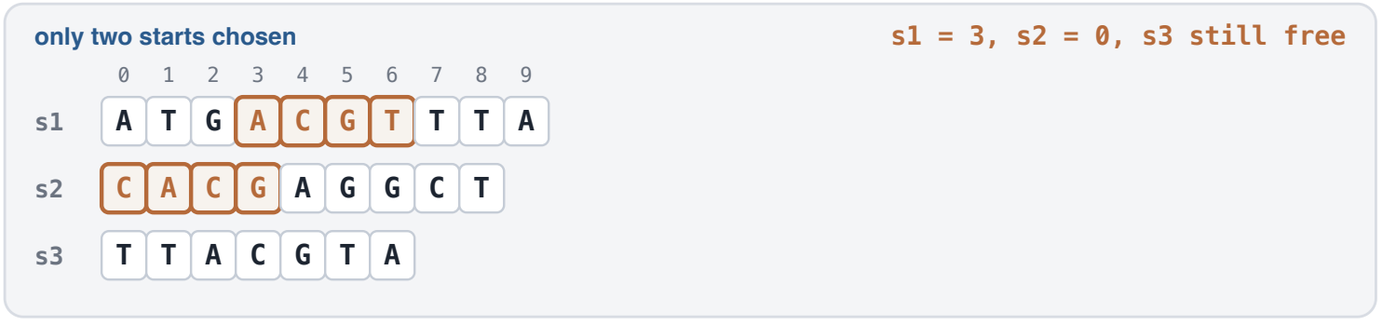
one mismatch, column 4

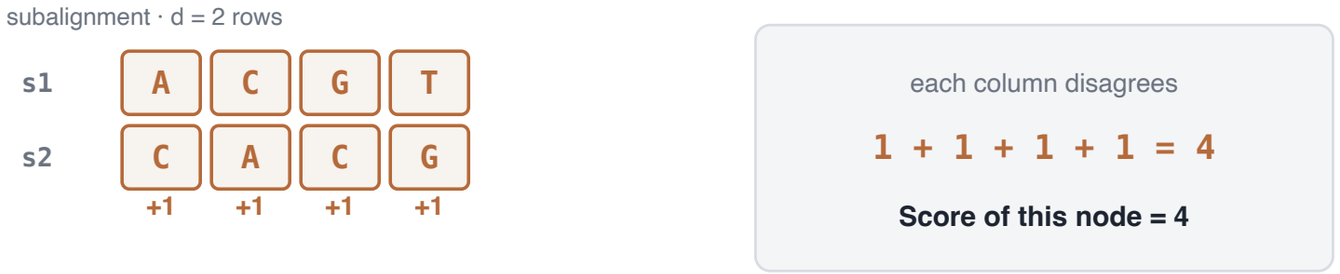
0 + 1 + 0 = 1

hold this as B

Score a subalignment

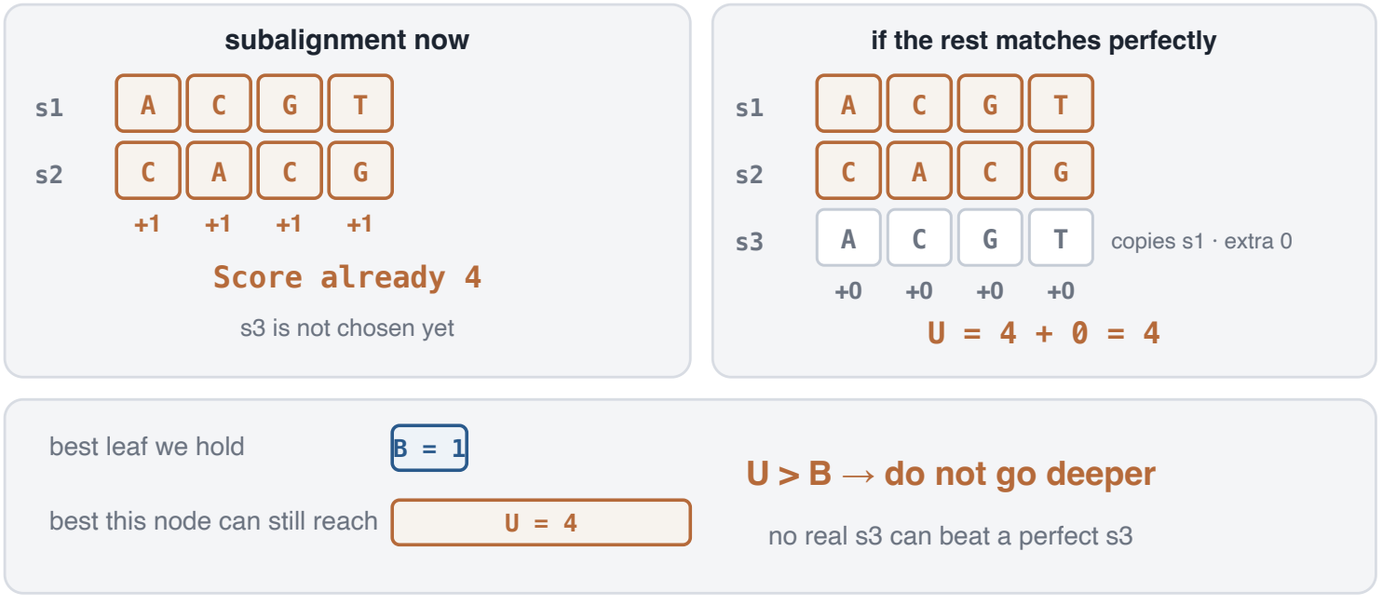
Only two starts are chosen: $s_1 = 3$ (`ACGT`) and $s_2 = 0$ (`CACG`). Each column whose letters differ already costs 1. That is $\text{Score} = 4$ on $d = 2$ rows.





Best possible: the rest matches perfectly

If unused rows copy a row we already have, they add 0. Then U equals the subalignment score — the best complete score this node can still reach.



> If $U > B$, even a perfect rest cannot beat the leaf we hold. Do not go deeper.

$U = 4$ is already worse than $B = 1$

$U = 4 > B = 1$. No start in s_3 can beat the leaf we hold. Skip every completion of this node.



do not go deeper: every s3 child is already worse than B = 1

the sibling with ACGA still has U = 1, so we keep walking there

still exact: every crossed-out leaf was already worse than the one we hold

The motif algorithm, in steps

$$\text{Score} = \sum_{c=1}^{k} \big(d - \max_{a} n_c(a)\big), \qquad U = \text{Score}$$

1. Walk the tree of starting positions. After the first complete leaf, hold its score as best-so-far B (on the toy, $B = 1$).
2. An edge picks a start in the next sequence. The windows chosen so far are a **subalignment** of d rows.
3. Score that subalignment. Remaining rows add 0 if they match perfectly, so U is the best this node can still reach.
4. If $U > B$, do not go deeper.
5. At a leaf, if $\text{Score} < B$, update B. The smallest B among surviving leaves is the motif score.

The notebook scores that bound

At a node, stack the windows you have. Per column: most frequent letter, then count the others. That sum is U. If $U > B$, prune.

```
B = 1                              # leaf s = (3, 1, 2)
U = score(["ACGT", "CACG"])        # columns 1+1+1+1 = 4
print(U > B)                       # True  →  prune
```

In the lab: confirm $U = 4$ prunes. Change the second window to `ACGA`. Does U still beat B?

Two bounds, one idea

	Restriction map	**Motif**
A node is	a partial set of sites X	starts chosen so far
An edge is	place from the left or from the right	a start in the next string
The bound	a required distance is **missing**	partial consensus score already $> B$

> Branch-and-bound is still exhaustive in the worst case (the tree may not prune). In good cases most subtrees die early.

90 seconds: which tree?

Pair: digest or motif?

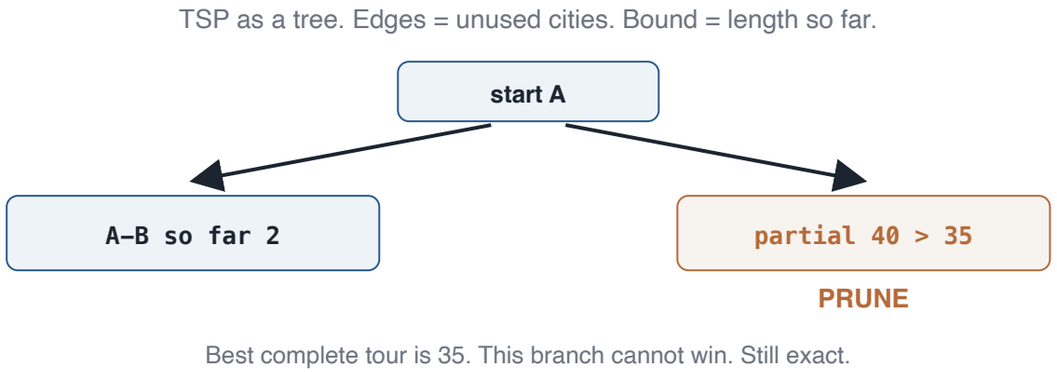
- The bound is Boolean (impossible vs possible).
- The bound is numeric (partial alignment score vs B).
- Branching factor 2. Branching factor $=$ starts in the next string.

The same prune on a tour

Week 2’s salesman, as a tree: unused cities on the edges. The bound is “length so far.” If that already beats the best complete tour, skip.

TSP as a search tree

Unused cities on the edges (week 2’s tour). Bound = length so far vs best complete tour.



Partial tour costs 40; best complete is 35

> **Q3** A partial tour already costs 40. You hold a complete tour of 35. What do you do with that branch, and why is the skip still exact?

This branch cannot win. **Prune.** Every skipped tour was already longer than one you hold. The bound is the length so far.

> Three labels, one skip: a missing distance, an optimistic Hamming score, a tour already longer than B.

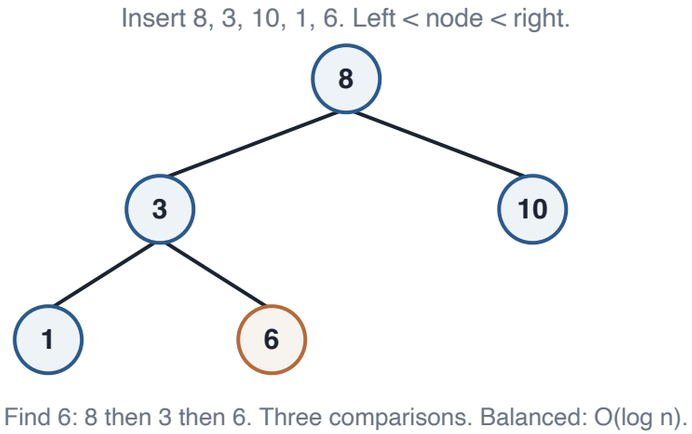
A tree that stores keys

So far a tree **searched candidates**. A binary search tree *stores* keys so each comparison can throw away half of them.

A different tree: store keys

So far a tree **searched candidates**. A **binary search tree** stores keys: left subtree $<$ node $<$ right subtree.

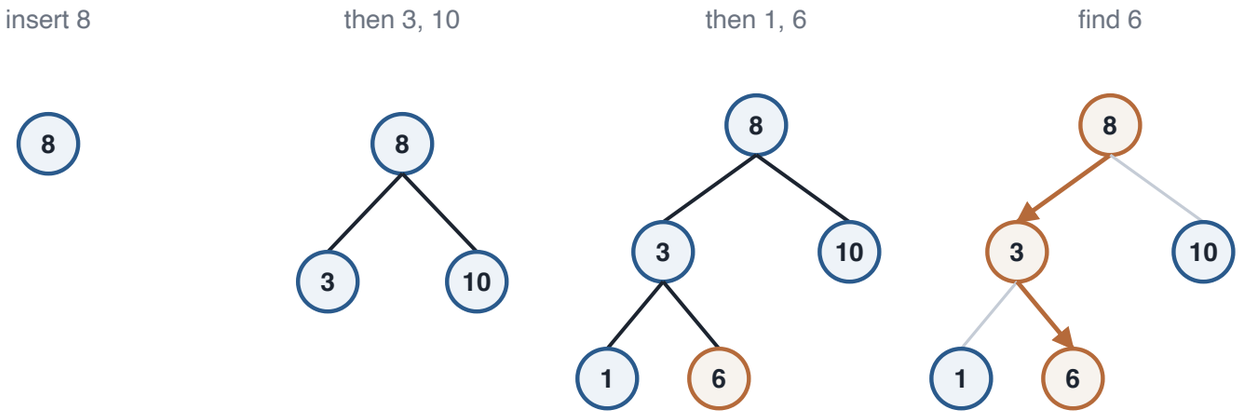
Search, insert: follow one child at each step.



Find 6: $8 \to 3 \to 6$. Three comparisons.

Grow it, then find 6

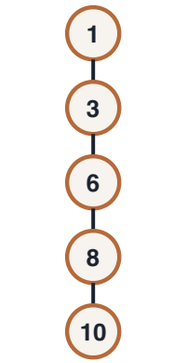
Insert 8, then 3 and 10, then 1 and 6. Left of a node is smaller, right is larger.



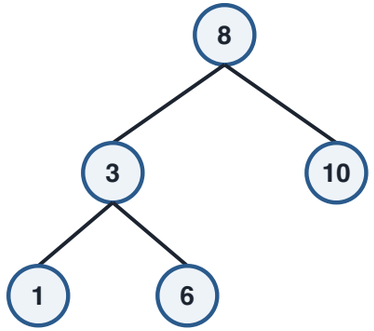
Left of a node is smaller, right is larger. Find 6: 8 → 3 → 6. Three comparisons.

Stick vs fan

Insert already-sorted keys $1, 3, 6, 8, 10$: each new key is larger, so it hangs on the right. A **stick** of height 4. Find is $O(n)$. Same five keys as a fan: height 2.



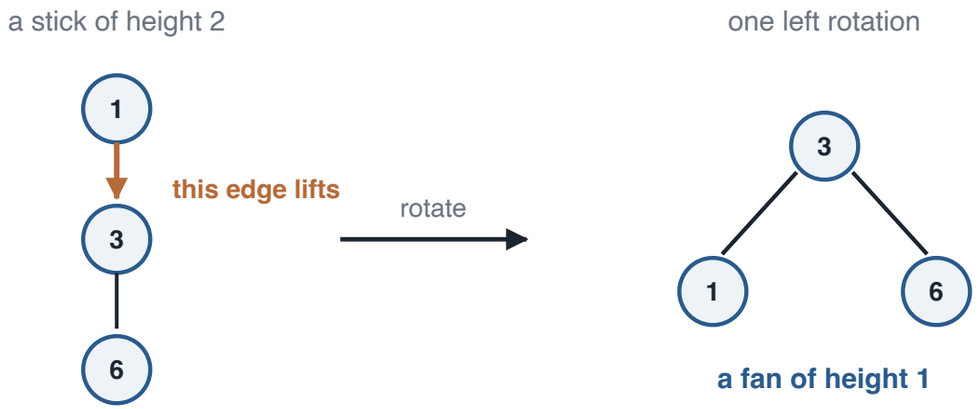
stick, height 4



fan, height 2

One rotation turns a stick into a fan

A three-node right spine has height 2. A **left rotation** on the middle edge lifts that edge: the middle key becomes the root, and the two sides become its children. Height 1.



Rotations exist so insert-sorted keys do not stay a stick.

AVL trees and red-black trees rotate after insert so the height stays logarithmic.

Cost is the height

If the tree is **balanced**, height $\Theta(\log n)$, so find/insert are $O(\log n)$.

If you insert already-sorted keys, the tree becomes a stick: height n, find is $O(n)$.

> The data structure is the idea: **order the keys in a tree** so each comparison halves the search (when the tree is balanced).

Open `labs/05-search.ipynb`

LAB · ~15 MINUTES · PYTHON

Select Kernel (Python 3). **Run All** (needs `matplotlib`). The lecture toys draw themselves: the cut map, then the DNA and the alignment.

1. **Digest.** `pairwise([0,2,4,7,10])` must equal L. Then read `legal([0,8,10], 7, L)` — it returns `(False, 1)`. That missing 1 is the bound.
2. **Play the other child.** Uncomment `legal([0, 2, 10], 7, L)`. Does that branch live? Why?
3. **Motif.** Leaf $s = (3, 1, 2)$ has $B = 1$. Partial `ACGT` / `CACG` has $U = 4 > B$, so prune. Change the second window to `ACGA`. Print U. Does this node die?
4. **Write Q1–Q3** in the notebook, without notes.

A bit of history

- **Land & Doig** (1960): branch-and-bound for integer programs. The prune is a proof that the subtree cannot succeed.
- **Little et al.** (1963): BnB for TSP. Week 2's tour list, as a tree.
- **Nathans & Smith** (1970s): restriction enzymes as a mapping tool — the biology that made today's distances a real experiment.
- **Skiena:** partial digest as a reconstruction puzzle — the toy is from that tradition.
- **Bailey & Elkan, MEME** (1994): motif finding in practice is often EM / Gibbs. The search picture is still a tree; the software usually refuses to walk all of it.

Tools that use this

- **Gurobi**, **CPLEX**, SCIP: industrial BnB.
- Phylogenetic **branch-and-bound** (exact parsimony on small n) in PAUP* and friends — week 10's tree space.
- **SAT** solvers: unit propagation is a bound by another name.
- Motif tools you will meet in a methods course: **MEME**, **HOMER**, **FIMO** — different algorithms, same biological question (a short noisy word).
- Databases and language runtimes: **balanced binary search trees** / B-trees, for lookups such as a gene name.
- VisuAlgo BST

Extensions

- **A***: BnB with a heuristic bound that is still a proof if the heuristic never overestimates.
- Double-digest maps (two enzymes): more lengths, still a reconstruction. We stay with one enzyme.
- AVL / red-black: the four rotation cases that keep a binary search tree logarithmic.

What you must be able to say

Without notes. Q1–Q2 today's idea. **Q3** a situation we did not drill.

Q1 Brute force as a tree: what sits on a node, what on an edge? (Digest *or* motif — say which.)

Q2 A bound that lets you skip a subtree — in one sentence. When does the whole tree still get explored?

Q3 Week 2's TSP: a partial tour already costs 40; you hold a complete tour of cost 35. What do you do with that branch, and why is that still exact?

Q3 asks about a situation we did not drill today.

Takeaways

A restriction map is recovered by walking a **tree of placements**. A missing distance kills a branch — that is branch-and-bound.

Motifs: the tree is over **starting positions**. A partial alignment whose consensus score already beats B dies.

A balanced binary search tree makes lookup logarithmic: the tree stores keys, and each comparison halves the remaining set.

Next week

Greedy algorithms and approximation.

Nearest neighbor was greedy and $O(n^2)$. Next: pancake flipping / reversals, and a number that says how bad greedy is allowed to be — the **approximation ratio**.

To read

- Land & Doig (1960). Branch-and-bound (integer programs; the same prune idea).
- Little, Murty, Sweeney, Karel (1963). An algorithm for the traveling salesman problem.
- Skiena, *The Algorithm Design Manual* — partial digest as a reconstruction puzzle.
- Bailey & Elkan (1994). Fitting a mixture model by expectation maximization to discover motifs in biopolymers. *ISMB* — what motif software actually runs.
- VisuAlgo — BST.